Reproducibility Audit Report: DelphiAgent (Xiong et al., 2025)

Session 6 deliverable · Research Methods & Scientific Integrity in AI · Doctoral Program in Deep Tech, UNMSM

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1. Paper metadata

Field	Entry
Title	DelphiAgent: A trustworthy multi-agent verification framework for automated fact verification
Authors	Xiong, C.; Zheng, G.; Ma, X.; Li, C.; Zeng, J.
Venue	Information Processing & Management, 62(6), 104241 (2025)
DOI	10.1016/j.ipm.2025.104241
Code repository	https://github.com/zjfgh2015/DelphiAgent (linked in the abstract)
Task	Automated fact verification of news claims with multiple LLM-based agents that emulate the Delphi method
Datasets named in the abstract	RAWFC and LIAR-RAW
Headline claim in the abstract	Macro-F1 improvements of up to 6.84% on RAWFC and comparable performance on LIAR-RAW, without a training regime
Why this paper	It belongs to the auditor's own field (LLM multi-agent verification of claims) and publishes code, so it can be run

2. Reproducibility scorecard

Each answer cites a page of the paper or a file of the repository. Items the paper does not report are marked Not reported, never guessed.

Item	What to look for	Y/ P/N	Evidence (page or file)
Seeds reported, many runs?	Mean ± SD, not one number	N	Paper: no seed, number of runs or dispersion is reported anywhere in the 17 pages; every cell of Tables 8 and 9 is a single number (pp. 9–10). Repo: a search of all .py files finds seed only in config.py (cache_seed = 42, a cache key), no temperature and no random.* call; each claim is evaluated once
Splits described?	Sizes, method,	Y	Table 7 and p. 8: RAWFC 1,589 / 200 / 200 and LIAR-RAW 10,065 / 1,274 / 1,251 (train / validation / test),

Item	What to look for	Y/ P/N	Evidence (page or file)
	grouped or time-based if needed		taken from Yang et al. (2022); “all empirical results are calculated on the same test sets” (p. 8). The paper does not describe how those splits were made
Leakage checked?	Preprocessing inside folds; no future features	P	Contamination risk is acknowledged (p. 8, section 4.3, p. 10) and WeiboEval (187 claims, posts after October 2023) was built to avoid it (p. 8). But RAWFC and LIAR-RAW, the main benchmarks, remain exposed, and WeiboEval is compared only with HiSS and ReAct (p. 10)
Tuning separated?	Validation set or nested CV	P	Validation splits exist (p. 8) and DelphiAgent needs no training (p. 8), but the choice of 3 rounds for RAWFC and 4 for LIAR-RAW, and of the prompts, is not tied to the validation set (p. 8)
Statistical test named?	Appropriate, paired when comparing models	N	Statistical significance is claimed against ReAct and HiSS (p. 9) without naming a test or giving a p-value; no test, p-value or bootstrap appears in the paper
Effect size and CI?	Not just a p-value	P	Gains are given as absolute differences, for example +6.84 macF1 over L-Defense (ChatGPT) on RAWFC (p. 9). No confidence interval or other uncertainty measure anywhere
External validation?	New site, new period or new population	P	WeiboEval is a new, author-built Chinese dataset from after the model cut-off (p. 8, Fig. 3 on p. 10): a different language and period, but only 187 claims and not independent
Compute documented?	Hardware, runtime, versions	N	Paper: no hardware, runtime, cost per claim, API dates, temperature or AutoGen version (AutoGen is only cited, p. 8; gpt-4o-2024-08-06 appears only in the reference list, p. 17). Repo: no requirements file or Python version; the README command installed autogen-agentchat 0.7.5 on Colab (Python 3.13.16) and the 0.2 series that the code needs could not be installed there
Code and data available?	A stranger could re-run it	P	Code and all 14 prompt files are public under Apache-2.0 (link on p. 1; prompt folder). The data availability statement says “Data will be made available on request” (p. 15). RAWFC and LIAR-RAW are only pointed to in another repository. WeiboEval.zip is in datasets/ (the README says a dataset directory), after repeated add and delete commits. The three scripts accept GuardEval and RAWFC only, with no LIAR-RAW option, and there is

Item	What to look for	Y/P/N	Evidence (page or file)
			no evaluation script. Both runs of get_predition.py failed on import (see section 7)

3. Experimental setup

From the repository (read on 25 October 2026): the system is built with AutoGen 0.2 and has three fact-checker agents with distinct personas (prompts Fact_Checker_P, M and N), a Synthesizer and a Finalizer. Fact checkers judge the claim and its verified evidence, the Synthesizer reports disagreements, and the cycle repeats up to three rounds before the Finalizer decides. The script get_predition.py takes –model_name (default gpt-4o-mini), –task (train, val or test, default test), –dataset (GuardEval or RAWFC) and writes one JSON file per claim. Input files carry event_id, claim, label and verified_evidence, which earlier scripts (evidence_extraction.py and evidence_verifier.py) are meant to produce.

From the paper (section 4.1, pp. 7–8): the datasets are RAWFC (Snopes) and LIAR-RAW (PolitiFact), created by Yang et al. (2022), plus the authors' own WeiboEval. Evidence mining uses gpt-4o-mini; the Delphi decision module is run with gpt-4o-mini and with gpt-4o. The maximum number of rounds is 3 for RAWFC and 4 for LIAR-RAW, the system is built with AutoGen, and the metrics are macro-averaged precision, recall and F1. Baseline numbers for L-Defense, RAFTS, ReAct and HiSS are copied from other papers, as the note to Table 8 says (p. 9). The paper reports no temperature, seed, number of runs, hardware or runtime.

4. Leakage check

The paper discusses a different kind of leakage: possible overlap between the benchmarks and the LLMs' training data (pp. 8 and 10). Its answer, WeiboEval, is a good practice but small (187 claims) and only compared with two baselines. There is no discussion of whether prompts, the number of rounds or the evidence-mining settings were chosen using the test sets. Because baseline results are taken from other papers (p. 9), the comparison does not guarantee identical conditions, such as the same LLM version or the same evidence.

5. Statistical rigor

All results are single runs, with no dispersion or confidence interval. The RAWFC test set has 200 claims, 66 or 67 per class (Table 7, p. 8), so one claim moves accuracy by half a point. As an illustration (auditor's calculation, not in the paper): for a proportion near 0.68 on 200 items, a 95% binomial interval is about ±6.5 points, similar in size to the headline gain of 6.84 macF1 points over the best supervised baseline (p. 9). A paired test on the same claims, or repeated runs, would be needed to judge that gain. The text claims "statistically significant" improvements over ReAct and HiSS (p. 9) without a test. On LIAR-RAW, DelphiAgent scores below L-Defense and FactLLaMA, which the paper acknowledges (p. 10).

6. Missing information

From the paper and from the repository (README.md, config.py and get_predition.py were read in full; all .py files were searched for seed, temperature and random):

- No seed, temperature, number of runs, hardware, runtime, cost or software versions in the paper; no requirements file or Python version in the repository.
- The README install command (pip install autogen-agentchat~=0.2) no longer gives the version the code needs, and the 0.2 series requires Python below 3.13 while the Colab runtime had 3.13.16.
- The order of the pipeline (evidence extraction, evidence verification, prediction) and the arguments of each script are not explained in the README.
- The paper reports LIAR-RAW results with up to 4 rounds (p. 8), but all three scripts accept only GuardEval and RAWFC and get_predition.py fixes max_turns = 3 (lines 338 and 351).
- The paper says data are available on request (p. 15); the README says WeiboEval is in a dataset directory; the folder is datasets and holds WeiboEval.zip, while the code calls the dataset GuardEval. The commit history shows several add and delete cycles of WeiboEval.zip.
- No script to compute the macro-F1 values of Tables 8 and 9.
- In get_predition.py the helper functions that build the messages are called without the dataset argument, so they use their English RAWFC default even when the Chinese-prompt branch is selected.
- Baseline results are copied from other papers (p. 9), so a reader cannot re-create the comparison from this repository alone.

7. Repository execution log

Repository contents (cloned on 25 October 2026, last commit 1ad2325 “Delete datasets/weibo”): config.py, evidence_extraction.py, evidence_verifier.py, get_predition.py, PromptH.py, README.md, LICENSE (Apache-2.0), two images, a datasets folder with WeiboEval.zip, and a prompt folder with 14 prompt files (Evidence_Extraction, Evidence_Verifier, three Fact_Checker personas, Synthesizer and Finalizer, each in _en and _ch versions). There is no requirements, environment, setup or pyproject file.

Execution in Google Colab, 25 October 2026 at 10:17: Python 3.13.16, Linux x86_64, 2 CPUs, 12 GiB of memory, no GPU.

Attempt 1: the README command pip install “autogen-agentchat~=0.2” installed autogen-agentchat 0.7.5 and autogen-core 0.7.5 (with openai 2.54.0), not a 0.2 release, because ~=0.2 accepts any 0.x version from 0.2 upwards. The installation also replaced protobuf with 5.29.6 and pip reported conflicts with google-cloud and tensorflow packages that Colab preinstalls, and asked for a runtime restart. Then import autogen failed with ModuleNotFoundError: No module named ‘autogen’. The scripts begin with from autogen import AssistantAgent, UserProxyAgent, so the install command given in the README no longer produces a working environment.

Attempt 2: pip install “autogen-agentchat>=0.2,<0.3” (the 0.2 series that the scripts need) failed with “No matching distribution found”. pip listed versions 0.2.36 to 0.2.40 as ignored because they require Python >=3.8 and <3.13, which means the Colab runtime’s Python is newer than 3.12. The only versions available for that Python are 0.4 and later, whose package layout differs

from the code. So on the default Colab runtime, the repository cannot be installed as published, and the README does not state a Python version.

Attempt 3 (searches over all .py files): the word seed appears only in config.py, as cache_seed = 42 for each of three model entries; there is no temperature setting and no call to random.* in any script. All three scripts (evidence_extraction.py, evidence_verifier.py, get_predition.py) accept only the datasets GuardEval and RAWFC, and get_predition.py sets max_turns = 3 in both branches.

Attempt 4 (runs): python get_predition.py –help and python get_predition.py –dataset RAWFC both stopped at line 3, from autogen import AssistantAgent, UserProxyAgent, ConversableAgent, with ModuleNotFoundError: No module named 'autogen'. No API key was used and no claim was processed.

Attempt 5 (Python 3.12 environment): a virtual environment with Python 3.12.3 and autogen-agentchat 0.2.40 was created with uv. In it, import autogen works, and python get_predition.py –help prints its options (–model_name, –data_dir, –task, –dataset with the choices GuardEval and RAWFC, –output_dir); only a warning that flaml.automl is not available appears. Running python get_predition.py –dataset RAWFC then stopped at line 46 of load_dataset with FileNotFoundError: No such file or directory: 'results/evidence_verify/RAWFC/test'. The repository has no results folder: that input is meant to be produced by the earlier scripts (evidence_extraction.py and evidence_verifier.py), which need the RAWFC data from another repository and calls to an OpenAI model, so the run could not go further without an API key and the data.

Summary of the run: the code can be installed only on Python 3.12 or older with a version pin that the README does not give, and even then the first script cannot run because its input data is not provided. No claim was processed, and no number of Tables 8 or 9 could be reproduced. The full commands and outputs are in the execution log notebook (DelphiAgent_execution_log.ipynb).

8. Overall score and justification

Score: 3.5 out of 9 (Yes = 1, Partial = 0.5, No = 0; this tally is the auditor's convention). One item is Yes (splits), five are Partial (leakage, tuning, effect size, external validation, code and data) and three are No (seeds and runs, statistical test, compute documentation).

Justification: the paper is clear about its datasets, splits and architecture, and it takes contamination seriously with WeiboEval. But its headline claims rest on single runs on a 200-claim test set, with significance asserted and no test, interval or compute information. The public code does not match the paper (no LIAR-RAW option, 3 rounds instead of 4, no evaluation script). In a Colab test, the README install command gave an incompatible package version; the version the code needs installed only in a Python 3.12 environment; and the main script then stopped because its input data (results/evidence_verify) is not in the repository. A stranger could understand the method but could not reproduce Tables 8 and 9.

9. Three recommendations

1. Report mean ± SD (or a bootstrap interval) over at least five runs with temperature and model versions fixed and recorded, and name a paired test (for example McNemar or a paired bootstrap) against the best baseline on the RAWFC test claims.
2. Release a runnable package: pinned requirements and Python version, run order, LIAR-RAW and WeiboEval support with the 4-round setting, the evaluation script for Tables 8 and 9, and the data (or checksums) that the README promises.
3. Document compute: API dates, temperature, cost per claim, runtime and hardware; and re-run the main baselines under the same conditions, or state clearly that their numbers are copied from other papers.